ITA1412 – ETHICAL HACKING

EXPERIMENT NO. 6

Experiment Name

Windows Operating System Commands Execution: Traceroute, Ping, Ifconfig and Netstat

Aim

To execute and observe network diagnostic commands in the Windows command prompt, as described in the lab manual.

Commands

Commands listed in the lab manual:

Command / Flag	Purpose	Example / Command
tracert	Trace the route to a host	tracert www.saveetha.com
ping	Test connectivity to an IP address	ping <IP address>
ifconfig	Command written in the manual	ifconfig
netstat	Display network connection information	netstat

Procedure

Step 1: Open Windows Command Prompt.

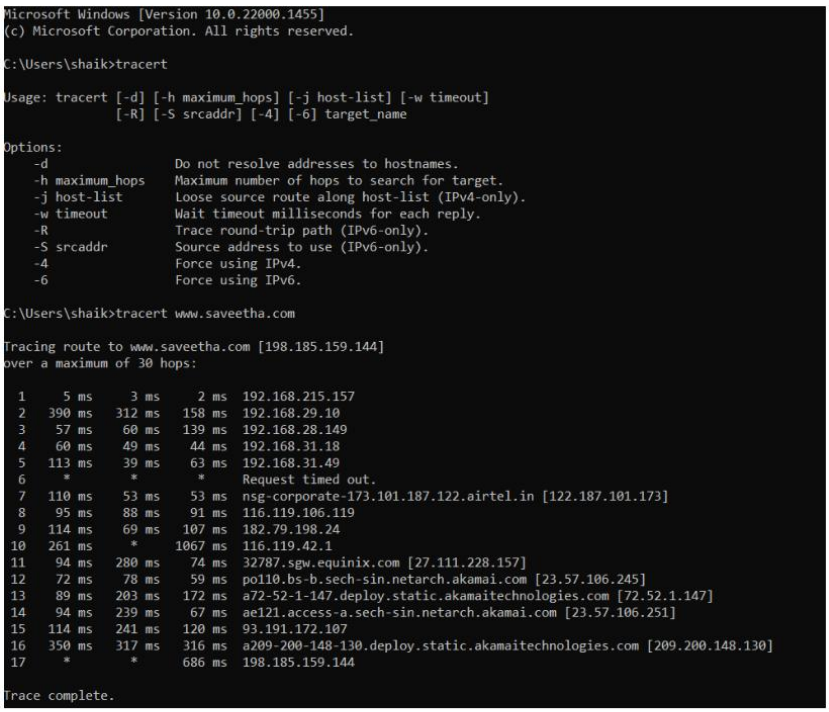
Step 2: Type tracert www.saveetha.com and press Enter.

Step 3: Type ping followed by an IP address and press Enter.

Step 4: Type ifconfig.

Step 5: Type netstat.

Screenshots



```
Microsoft Windows [Version 10.0.22000.1455]
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C:\Users\shaik>tracert

Usage: tracert [-d] [-h maximum_hops] [-j host-list] [-w timeout]
               [-R] [-S srcaddr] [-4] [-6] target_name

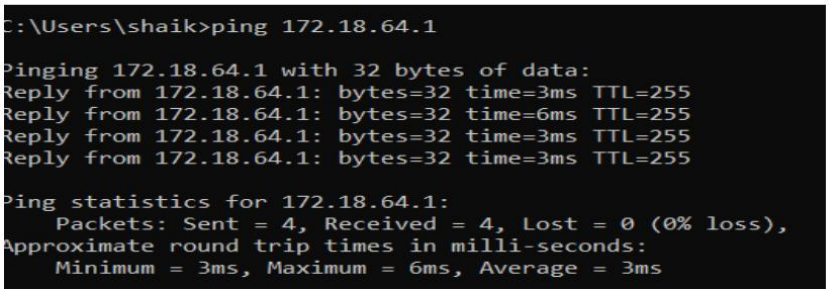
Options:
    -d                 Do not resolve addresses to hostnames.
    -h maximum_hops    Maximum number of hops to search for target.
    -j host-list       Loose source route along host-list (IPv4-only).
    -w timeout         Wait timeout milliseconds for each reply.
    -R                 Trace round-trip path (IPv6-only).
    -S srcaddr         Source address to use (IPv6-only).
    -4                 Force using IPv4.
    -6                 Force using IPv6.

C:\Users\shaik>tracert www.saveetha.com

Tracing route to www.saveetha.com [198.185.159.144]
over a maximum of 30 hops:

  1     5 ms     3 ms     2 ms  192.168.215.157
  2   390 ms   312 ms   158 ms  192.168.29.10
  3    57 ms    60 ms   139 ms  192.168.28.149
  4    60 ms    49 ms    44 ms  192.168.31.18
  5   113 ms    39 ms    63 ms  192.168.31.49
  6     *        *        *     Request timed out.
  7   110 ms    53 ms    53 ms  nsg-corporate-173.101.187.122.airtel.in [122.187.101.173]
  8    95 ms    88 ms    91 ms  116.119.106.119
  9   114 ms    69 ms   107 ms  182.79.198.24
 10   261 ms     *     1067 ms  116.119.42.1
 11    94 ms   280 ms    74 ms  32787.sgw.equinix.com [27.111.228.157]
 12    72 ms    78 ms    59 ms  po110.bs-b.sech-sin.netarch.akamai.com [23.57.106.245]
 13    89 ms   203 ms   172 ms  a72-52-1-147.deploy.static.akamaitechnologies.com [72.52.1.147]
 14    94 ms   239 ms    67 ms  ae121.access-a.sech-sin.netarch.akamai.com [23.57.106.251]
 15   114 ms   241 ms   120 ms  93.191.172.107
 16   350 ms   317 ms   316 ms  a209-200-148-130.deploy.static.akamaitechnologies.com [209.200.148.130]
 17     *        *      686 ms  198.185.159.144

Trace complete.
```



```
C:\Users\shaik>ping 172.18.64.1

Pinging 172.18.64.1 with 32 bytes of data:
Reply from 172.18.64.1: bytes=32 time=3ms TTL=255
Reply from 172.18.64.1: bytes=32 time=6ms TTL=255
Reply from 172.18.64.1: bytes=32 time=3ms TTL=255
Reply from 172.18.64.1: bytes=32 time=3ms TTL=255

Ping statistics for 172.18.64.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 3ms, Maximum = 6ms, Average = 3ms
```